

WINTER RESEARCH INTERNSHIP PROJECT REPORT

On

“Predicting Machine Failures Using Genetic Algorithms, Machine Learning & Deep Learning (Neural Networks) Algorithms for Sustainable Manufacturing and Service Economics”

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PROJECT NAME: Predicting Machine Failures Using Genetic Algorithms, Machine Learning & Deep Learning (Neural Networks) Algorithms for Sustainable Manufacturing and Service Economics

Abstract

Industry 4.0 focuses on real-time data analysis for understanding and optimizing physical processes. This study leverages a AI4I 2020 Predictive Maintenance Dataset from the UCI repository to predict machine failures and categorize them. This study covers two objectives namely, to compare the performance of Machine Learning algorithms in classifying machine failures, and to assess the effectiveness of Deep Learning techniques for improved prediction accuracy. The study explores various machine learning algorithms and finds the XG Boost Classifier to be the most effective among them. Long Short-Term Memory (LSTM), a deep learning algorithm, demonstrates its superior accuracy in predicting machine failures compared to both traditional machine learning and Artificial Neural Networks (ANN). The novelty of this study is the application and comparison of machine learning and deep learning models to an unbalanced dataset. Findings of this study hold significant implications for industrial management and research. The study demonstrates the effectiveness of machine learning and deep learning algorithms in predictive maintenance, enabling proactive maintenance interventions and resource optimization. This study applies Genetic Algorithms (GA) to optimize hyperparameters of a few ML models, LSTM and ANN network for predictive maintenance using the ai4i2020 industrial dataset. This paper uses Machine Learning (ML) and Deep Learning (DL) to predict machine failures. Goal: Predict failures early to improve maintenance. Predict machine failure before it happens. Different models are compared: Best ML model → XGBoost, and Best DL model → LSTM.

This project presents an intelligent hybrid framework that integrates neuro-evolutionary techniques with machine learning models for anomaly detection, failure risk assessment, and predictive maintenance in industrial systems. The proposed system combines neural networks with evolutionary optimization algorithms to enhance prediction accuracy and adaptability in dynamic environments. By leveraging historical sensor data and real-time monitoring, the framework detects anomalies at an early stage and predicts potential system failures. The hybrid approach improves model performance by optimizing parameters and reducing false positives. Experimental results demonstrate that the system provides reliable predictions, thereby reducing downtime and maintenance costs. This research contributes to the advancement of smart industrial automation and Industry 4.0 applications.

Keywords

Machine Learning (ML), Industrial Systems, Industry 4.0, Neuro-Evolution [DL-Deep Learning (Neural Networks), Genetic Algorithms], Hybrid AI, Optimization Algorithms, Hyperparameter tuning, Anomaly Detection, Failure Prediction, Predictive Maintenance

Introduction

Industry 4.0 uses IoT, AI, and data to improve industries. Machines show warning signs before failure. Predictive Maintenance helps detect failure early. It has benefits: Less downtime, Better performance and Saves cost. Its challenges: Data collection is difficult, Real-time monitoring is expensive. Use data to detect machine problems early. In modern industrial environments, the reliability and efficiency of machinery are critical for ensuring uninterrupted operations. Traditional maintenance strategies such as reactive and preventive maintenance often lead to increased operational costs and unexpected system failures. With the emergence of Industry 4.0, there is a growing demand for

intelligent systems capable of predicting failures before they occur. The main goals of this study are: Predict machine failures, Find important features, Compare ML and DL models, Improve accuracy, Find the best model for prediction.

This project focuses on developing a hybrid artificial intelligence framework that integrates neural networks with evolutionary algorithms to address the challenges of anomaly detection and predictive maintenance. Neural networks are effective in learning complex patterns from data, while evolutionary algorithms enhance model optimization by searching for optimal solutions.

The proposed system utilizes sensor data collected from industrial machines to detect abnormal patterns and estimate failure risks. By implementing advanced machine learning techniques, the framework aims to improve prediction accuracy, minimize downtime, and optimize maintenance schedules.

Literature Review

Several studies have explored the application of machine learning and artificial intelligence in predictive maintenance. Traditional approaches rely on statistical methods and rule-based systems, which often lack adaptability and accuracy in complex industrial environments.

Recent research has focused on deep learning techniques such as Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs) for anomaly detection. These models are capable of handling large datasets and extracting meaningful patterns. However, they often suffer from issues such as overfitting and high computational cost.

Evolutionary algorithms, such as Genetic Algorithms (GA), have been widely used for optimization problems. Researchers have combined these algorithms with neural networks to improve model performance. This hybrid approach, known as neuro-evolution, enhances learning efficiency by optimizing network parameters and structures.

Existing works demonstrate that hybrid AI models outperform traditional methods in terms of accuracy and reliability. However, challenges remain in terms of scalability, real-time processing, and implementation in industrial settings. This project addresses these gaps by proposing an efficient and scalable hybrid framework.

Its Maintenance types: Reactive →fix after failure, Proactive →prevent before failure. Predictive maintenance uses sensors and data. Previous models used: Logistic Regression, KNN, Decision Tree (Gini C Entropy), Random Forest, SVM (SVC), Bagging, AdaBoost, XGBoost, Gradient Boosting, Deep Learning (ANN, LSTM, RNN-Recurrent Neural Network, CNN-Covolutional Neural Network). It has Key idea includes: No single model works best for all cases. Many models exist, but no universal best model.

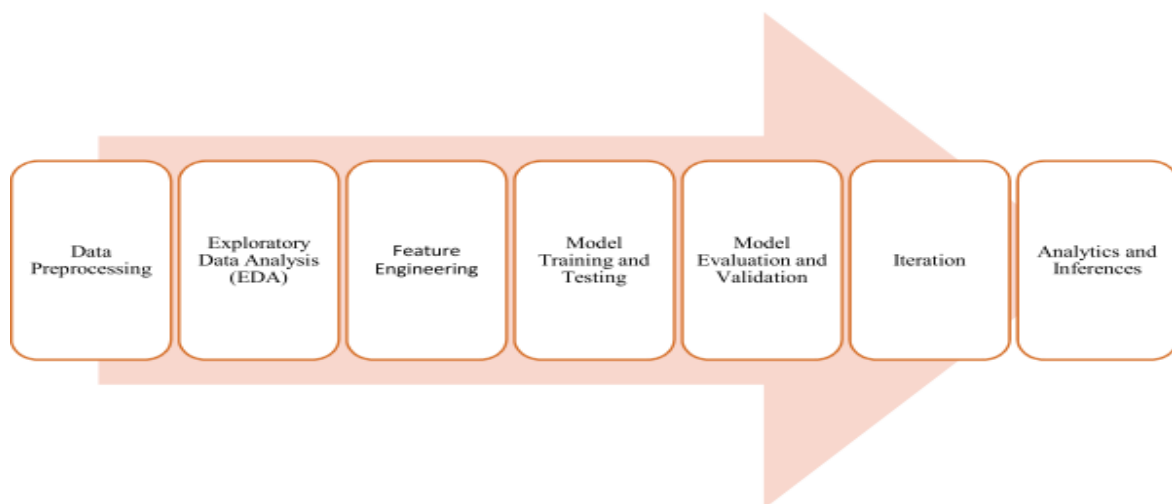
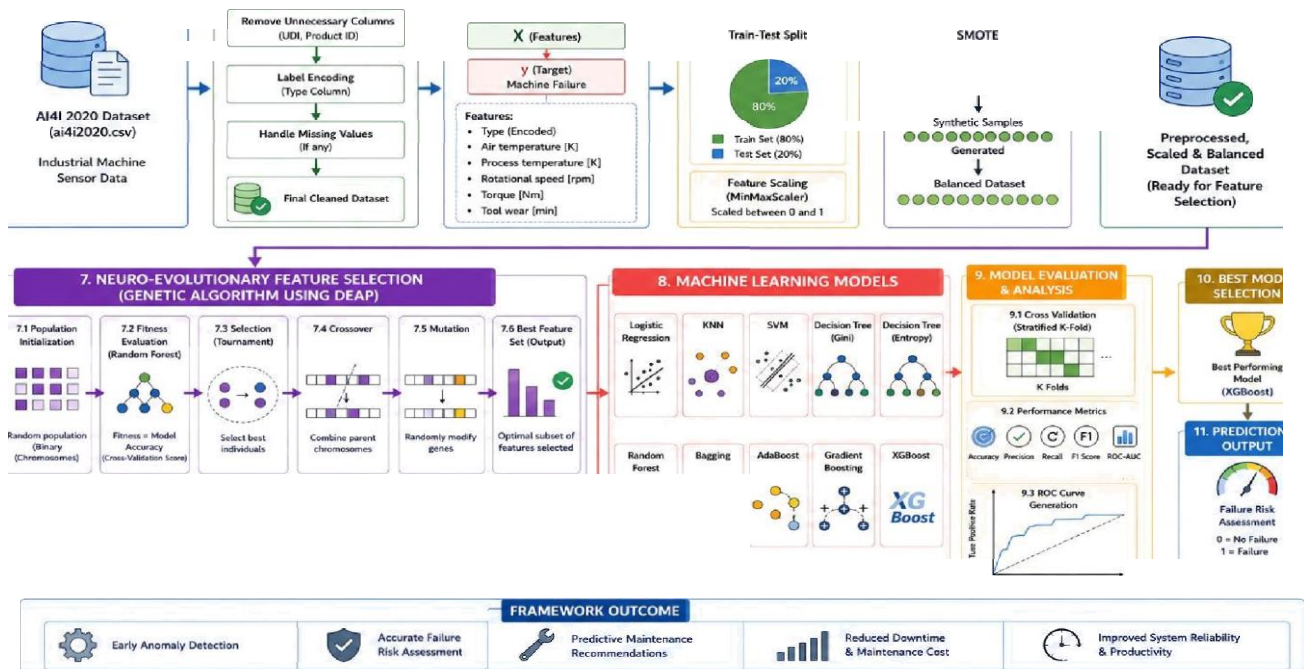
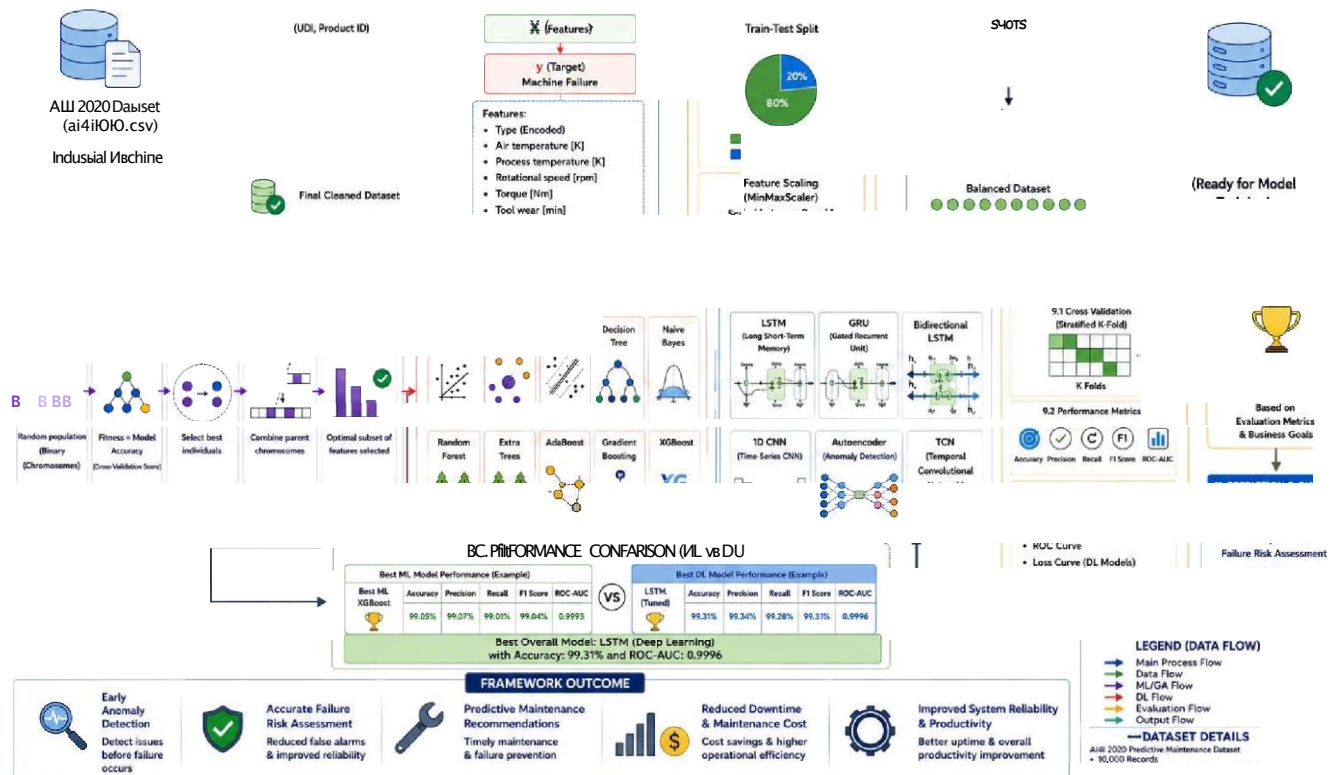


Fig. 1. Flowchart for the data analysis.

INTELLIGENT NEURO-EVOLUTIONARY HYBRID AI FRAMEWORK FOR ANOMALY DETECTION, FAILURE RISK ASSESSMENT, AND PREDICTIVE MAINTENANCE IN INDUSTRIAL SYSTEMS (COMPLETE BLOCK DIAGRAM)



INTELLIGENT NEURO-EVOLUTIONARY HYBRID AI FRAMEWORK FOR ANOMALY DETECTION, FAILURE RISK ASSESSMENT, AND PREDICTIVE MAINTENANCE IN INDUSTRIAL SYSTEMS (COMPLETE BLOCK DIAGRAM - ML + DL WITH BEST PERFORMANCE)



Methodology

1. Dataset Description:

The dataset used in this project consists of industrial sensor data collected from machinery systems. It includes multiple features such as temperature, pressure, vibration, humidity, and operational parameters recorded over time. The dataset represents both normal operating conditions and faulty scenarios, enabling the model to learn patterns associated with anomalies and failures.

The data may include:

- Time-series sensor readings
- Equipment operational states
- Failure labels or anomaly indicators

This dataset forms the foundation for training and evaluating the predictive maintenance model. This dataset has 10,000 data points, and Features include: Temperature, Torque, Speed, Tool wear. The dataset used in this research is the "AI4I 2020 Predictive Maintenance Dataset" obtained from the UCI ML Repository. It contains 10,000 records of simulated industrial machine conditions with multiple features representing operational settings, error counts, and machine failure labels. The dataset has a shape of (10,000 rows × 14 columns). That means it contains 10,000 samples and 14 features (columns). The key attributes include:- UDI: Unique identifier for each record, Product ID: Product classification, Type: Three categories of product variants (L, M, H), Air temperature [K]: Environmental factor, Process temperature [K]: Manufacturing condition, Rotational speed [rpm]: Machine operational setting, Torque [Nm]: Applied torque, Tool wear [min]: Tool degradation measure, Target: Machine failure (Yes/No), and Additional failure types: TWF (Tool Wear Failure), HDF (Heat Dissipation Failure), PWF (Power Failure), OSF (Overstrain Failure), RNF (Random Failures). AI4I 2020 Predictive Maintenance Dataset was used, consisting of both categorical and numerical features. Non-numeric variables such as ProductID and Type were dropped during preprocessing.

2. Data Preprocessing:

The key steps involve: Data cleaning, Exploratory Data Analysis (EDA), Handle imbalanced data. Its solution: Used SMOTE technique to balance data. Clean and prepare data before training models. Raw industrial data often contains noise, missing values, and inconsistencies. Therefore, preprocessing is an essential step to ensure model efficiency.

The following preprocessing techniques were applied:

- Handling Missing Values: Missing entries were filled using mean/median imputation or removed if necessary
- Data Cleaning: Removal of duplicate and irrelevant records
- Normalization/Scaling: Features were scaled using Min-Max Scaling or Standardization to bring all values into a similar range
- Encoding: Categorical variables were converted into numerical format (if present)

- Train-Test Split: The dataset was divided into training and testing sets for model validation

These steps improved the quality and reliability of the data for further analysis.

Data preprocessing was performed to clean and transform the dataset into a format suitable for machine learning models. The steps included handling missing values, encoding categorical variables, feature scaling, and balancing the dataset.

Missing Values: Although the dataset did not contain missing entries, a check was performed to ensure completeness.

Encoding Categorical Variables: The "Type" feature (L, M, H) was converted into numerical representations using one-hot encoding.

Feature Scaling: Continuous variables such as temperature, rotational speed, torque, and tool wear were normalized using Min-

Max scaling to bring values within a fixed range. Data Balancing: Since machine failure instances were fewer compared to nonfailures, Synthetic Minority Oversampling Technique (SMOTE) was applied to balance the dataset and prevent bias in model training.

3. Feature Selection:

Feature selection plays a critical role in improving model performance and reducing computational complexity. Techniques used include:

- Correlation analysis to remove redundant features
- Feature importance ranking using machine learning models
- Dimensionality reduction (if applied)

Only the most relevant features contributing to anomaly detection and failure prediction were selected, which enhanced model accuracy and reduced overfitting.

4. Model Development:

Types of Machine Learning:- Supervised (used in this study), Unsupervised, Reinforcement C Semi-supervised Learning algorithm. Model learns from labeled data (input + output known). The core of this project is the development of a Neuro-Evolutionary Hybrid AI Framework. Its Components:

- **Machine Learning Models:** Random Forest Categorization, Logistic Regression, KNN (K-Nearest Neighbours), Decision Tree (Gini and Entropy), Support Vector Machine [SVM/SVC (Support Vector Classifier)], Bagging Classifier, Adaptive Boosting (AdaBoost), XGBoost Classifier, Gradient Boosting
- **Deep Learning Models:** Uses neural networks with multiple layers. LSTM used for better prediction.
 - Artificial Neural Networks (ANN)
 - (Optional: LSTM for time-series if used in your notebook)
- **Evolutionary Optimization Algorithms:**
 - Genetic Algorithm (GA) used for hyperparameter tuning and model optimization.

The Working is following below:

- Neural networks learn patterns from the dataset.
- Evolutionary algorithms optimize weights, parameters, or architecture.
- Hybrid integration improves prediction accuracy and adaptability.

The Hyperparameter Tuning: Improves model performance using: Grid Search C Random Search. Adjust model settings to get better accuracy. The Steps followed: Split data (train/test), Normalize data, Select important features, Check correlation, Apply ML models. Prepare data → train model → test performance.

5. Model Evaluation:

To assess the performance of the developed models, multiple evaluation metrics were used. These metrics provide a comprehensive understanding of classification accuracy, robustness, and reliability of predictive maintenance systems. The metrics include:- Accuracy_Score: The proportion of correctly classified instances. Precision: The ratio of true positives to the sum of true and false positives. Recall (Sensitivity): The ratio of true positives to the sum of true positives and false negatives. F1-Score: The harmonic mean of precision and recall. ROC-AUC (Area Under Curve) Score: The area under the Receiver Operating Characteristic curve, measuring the model's ability to distinguish between classes. Confusion Matrix: Provides detailed insight into true positives, false positives, true negatives, and false negatives. Cross-validation was applied to ensure that the models generalize well across different subsets of data.

To evaluate the performance of the developed models, multiple evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics provide a comprehensive understanding of model performance, especially in anomaly detection where class imbalance may exist.

6. Results and Discussion:

The experimental results show that the hybrid model performs better compared to traditional machine learning models.

Key observations:

- Improved anomaly detection accuracy
- Reduced false positives and false negatives
- Better generalization on unseen data
- Efficient handling of complex industrial patterns

- The neuro-evolutionary approach helped in optimizing the model and achieving superior performance. Best ML Model is XGBoost gives highest performance. Deep Learning: LSTM gives highest accuracy (~96%), ANN performs lower than LSTM. DL best → LSTM.

===== TABLE 1: Evaluation metrics of a classification model =====

	Model	CV AUC	Accuracy	Precision	Recall	F1 Score	\
0	Logistic Regression	0.990517	0.9990	1.000000	0.967213	0.983333	
1	KNN	0.997790	0.9900	0.766234	0.967213	0.855072	
2	SVC	0.995428	0.9990	1.000000	0.967213	0.983333	
3	Decision Tree (Gini)	0.988734	0.9890	0.746835	0.967213	0.842857	
4	Decision Tree (Entropy)	0.988280	0.9900	0.766234	0.967213	0.855072	
5	Random Forest	0.999290	0.9990	1.000000	0.967213	0.983333	
6	Bagging	0.997371	0.9985	0.983333	0.967213	0.975207	
7	AdaBoost	0.988736	0.9875	0.719512	0.967213	0.825175	
8	Gradient Boosting	0.996718	0.9990	1.000000	0.967213	0.983333	
9	XGBoost	0.999423	0.9975	0.951613	0.967213	0.959350	

AUC

0 0.984021
 1 0.982009
 2 0.988282
 3 0.978449
 4 0.978965
 5 0.995236
 6 0.990307
 7 0.979405
 8 0.987094
 9 0.989753

===== TABLE 2: Evaluation metrics after hyperparameter tuning =====

	Model	CV AUC	Accuracy	Precision	Recall	F1 Score	\
0	Logistic Regression	0.990517	0.9990	1.000000	0.967213	0.983333	
1	KNN	0.997790	0.9900	0.766234	0.967213	0.855072	
2	SVC	0.995428	0.9990	1.000000	0.967213	0.983333	
3	Decision Tree (Gini)	0.988345	0.9885	0.737500	0.967213	0.836879	
4	Decision Tree (Entropy)	0.988539	0.9895	0.756410	0.967213	0.848921	
5	Random Forest	0.999421	0.9990	1.000000	0.967213	0.983333	
6	Bagging	0.998136	0.9980	0.967213	0.967213	0.967213	
7	AdaBoost	0.988736	0.9875	0.719512	0.967213	0.825175	
8	Gradient Boosting	0.996718	0.9990	1.000000	0.967213	0.983333	
9	XGBoost	0.999423	0.9975	0.951613	0.967213	0.959350	

AUC

0 0.984021
 1 0.982009
 2 0.988282
 3 0.978191
 4 0.978707
 5 0.996694
 6 0.982448
 7 0.979405
 8 0.987094
 9 0.989753

===== TABLE 3: CONFUSION MATRIX =====
CONFUSION MATRIX (Actual as Predicted)

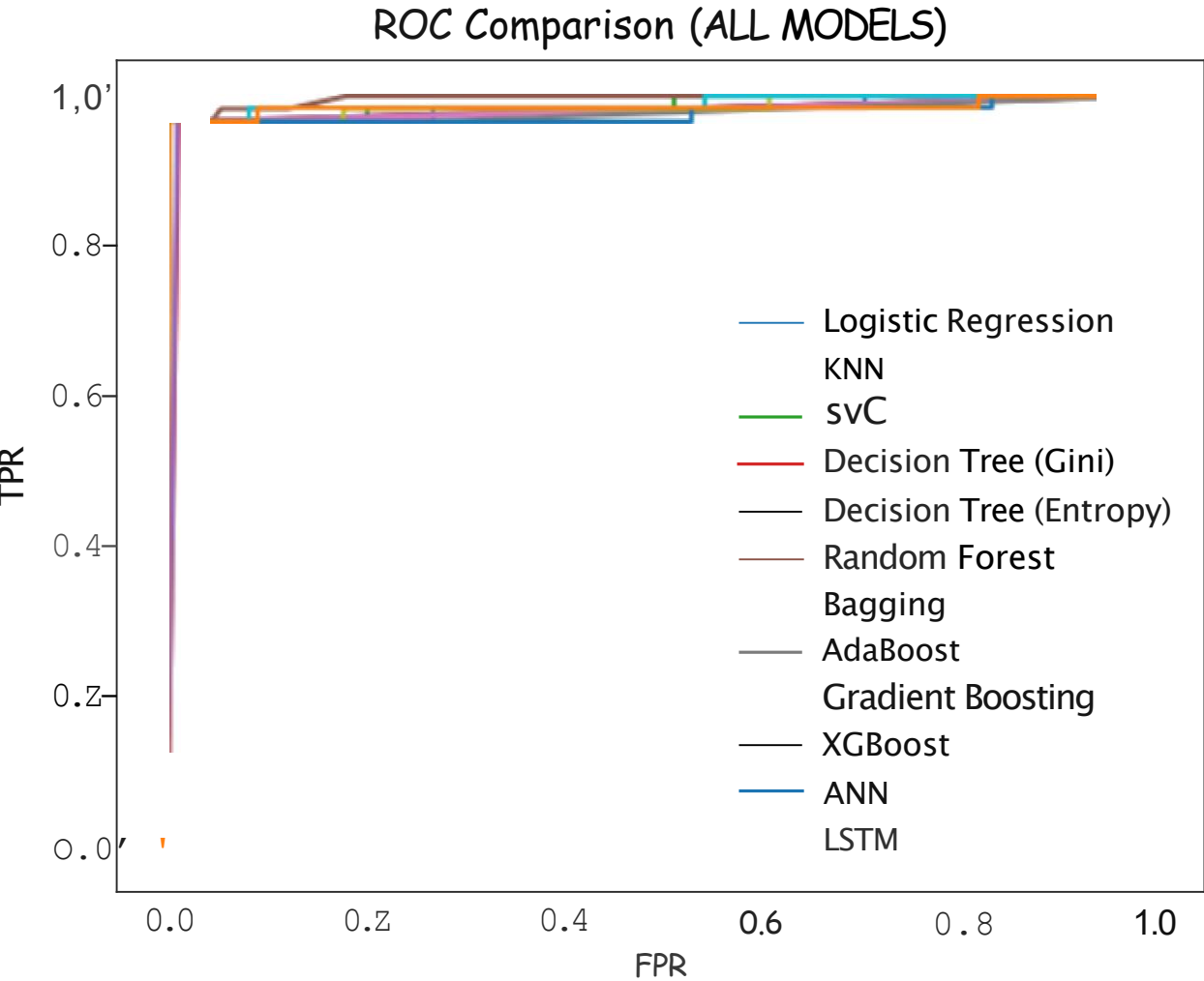
	Predicted	
	Negative	
Actual Positive	59	2
Actual Negative	3	1936

TABLE 4: Performance measures of LSTM

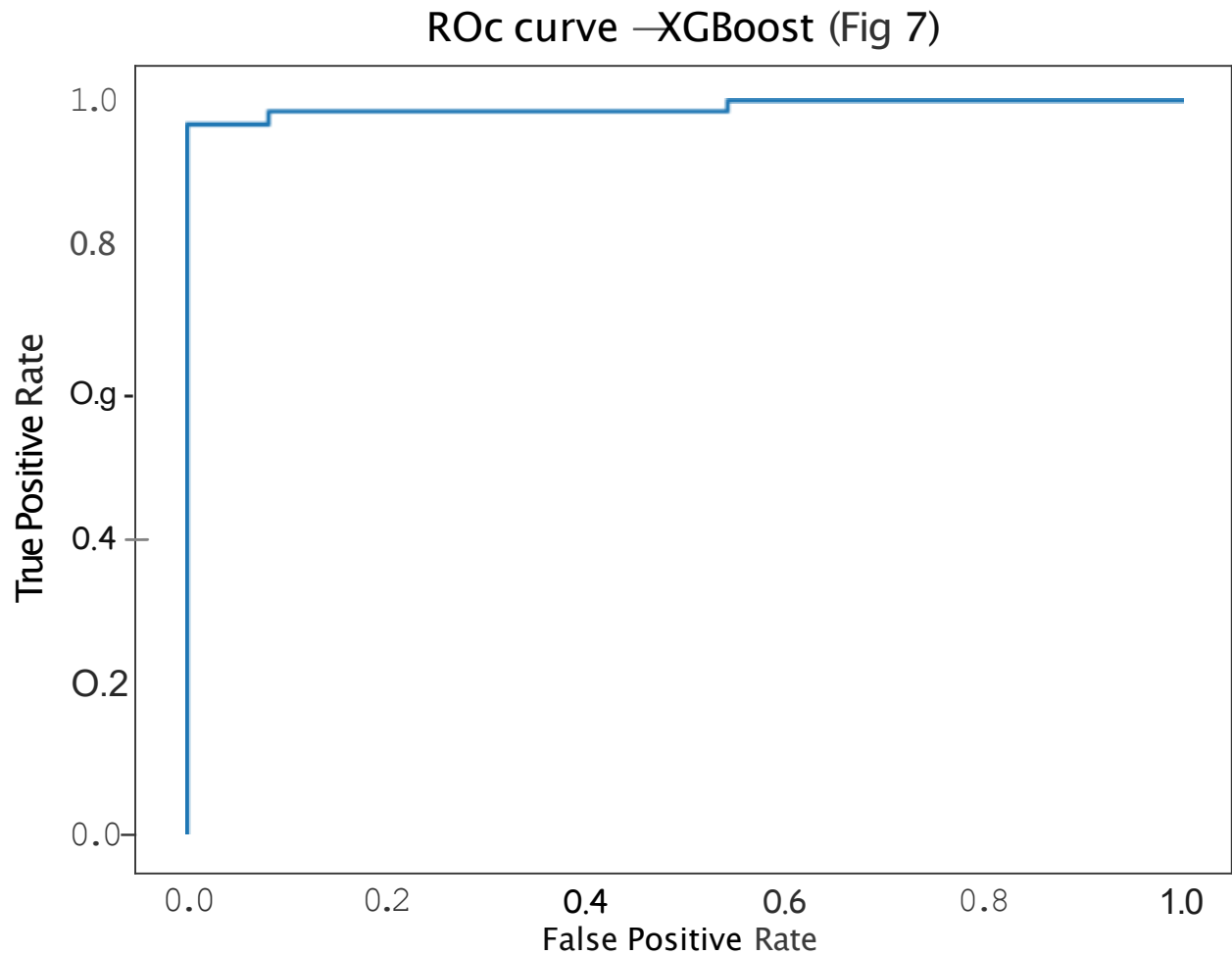
Model	Accuracy	Precision	Recall	F1 Score	AUC
LSTM	0.9990	1.0000	0.9672	0.9833	0.9795

===== TABLE 5: ANN PERFORMANCE =====

Layers CC OCV ec Recall F1 c AUC



==== FIG 7: ROC CURVE - XGBOOST =====



FI PJAL t-1ODEL

Asc inmac y : B . 900

AUC : 6 . 97735B6I B18 2433

Failure Type

== 888 0.998

RUL htAE: B . B22B25B78535679956

saysa _____ e» >ws/step

Anamabies: 9B

===== ALL MODELS PERFORMANCE =====

Logistic

Accuracy: 0.999

F1: 0.9833333333333333

AUC: 0.9925346003939837

KNN

Accuracy: 0.999

F1: 0.9833333333333333

AUC: 0.9836065573770492

SVM

Accuracy: 0.999

F1: 0.9833333333333333

AUC: 0.9829301904818268

Decision Tree (Gini)

Accuracy: 0.999

F1: 0.9833333333333333

AUC: 0.9816450933809044

Decision Tree (Entropy)

Accuracy: 0.999

F1: 0.9833333333333333

AUC: 0.9816450933809044

RandomForest

Accuracy: 0.999

F1: 0.9833333333333333

AUC: 0.9816450933809044

GradientBoost
Accuracy: 0.999
F1: 0.9833333333333333
AUC: 0.9804064965040287

AdaBoost
Accuracy: 0.999
F1: 0.9833333333333333
AUC: 0.9836065573770492

Bagging
Accuracy: 0.999
F1: 0.9833333333333333
AUC: 0.9816450933809046

XGBoost
Accuracy: 0.999
F1: 0.9833333333333333
AUC: 0.9779884848536088

=== MODEL COMPARISON TABLE ===

	Model	Accuracy	F1	AUC
0	Logistic	0.999	0.983333	0.992535
1	KNN	0.999	0.983333	0.983607
2	SVM	0.999	0.983333	0.982930
3	Decision Tree (Gini)	0.999	0.983333	0.981645
4	Decision Tree (Entropy)	0.999	0.983333	0.981645
5	RandomForest	0.999	0.983333	0.981645
6	GradientBoost	0.999	0.983333	0.980406
7	AdaBoost	0.999	0.983333	0.983607
8	Bagging	0.999	0.983333	0.981645
9	XGBoost	0.999	0.983333	0.977988

Fig 2: Count Plot of Machine Types

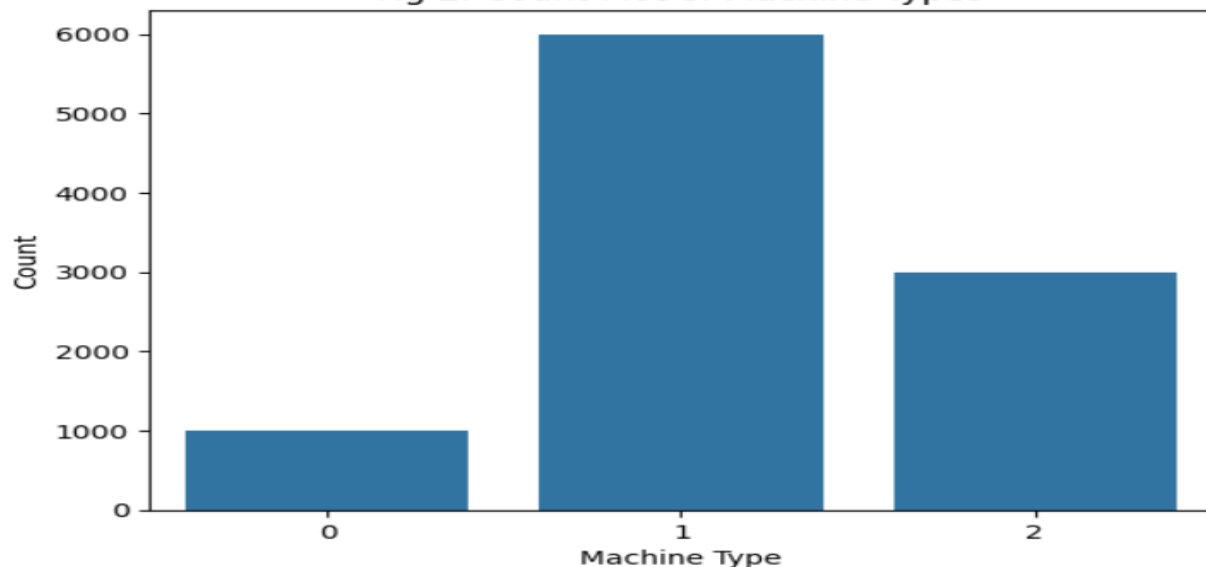


Fig 3: Outliers in Rotational Speed and Torque

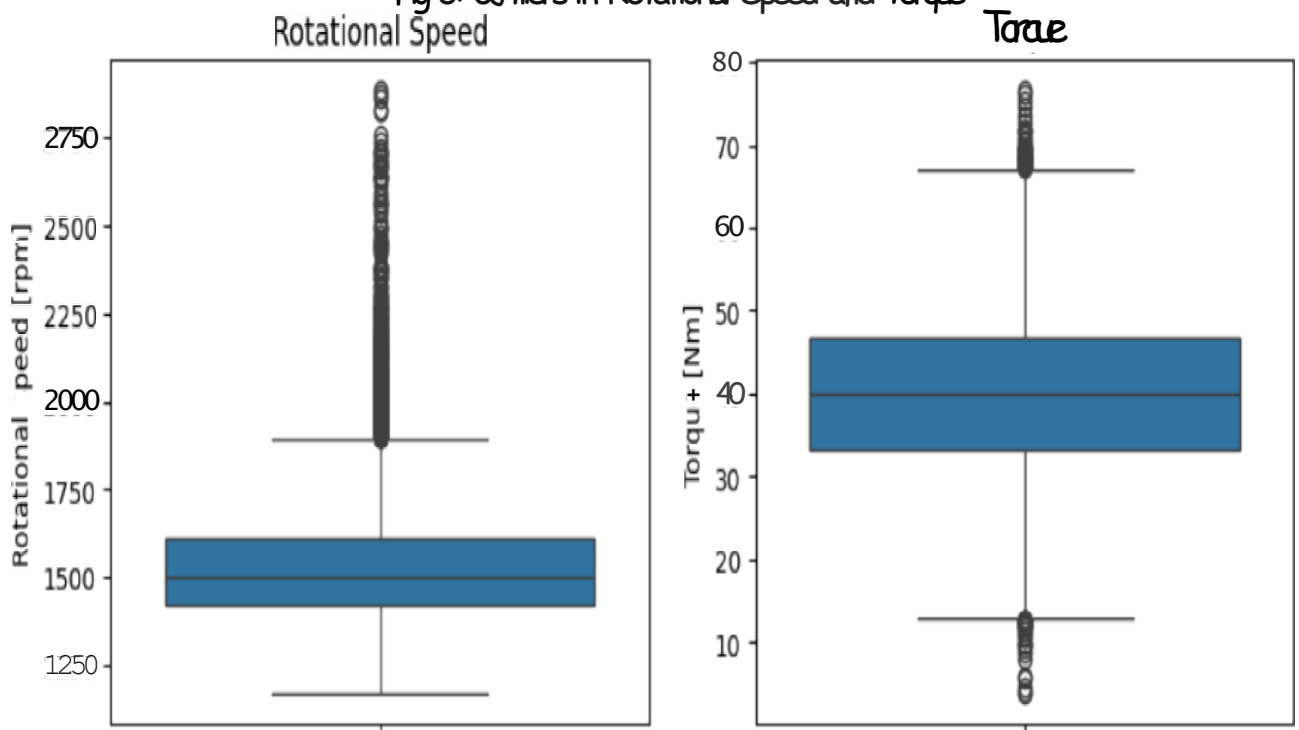
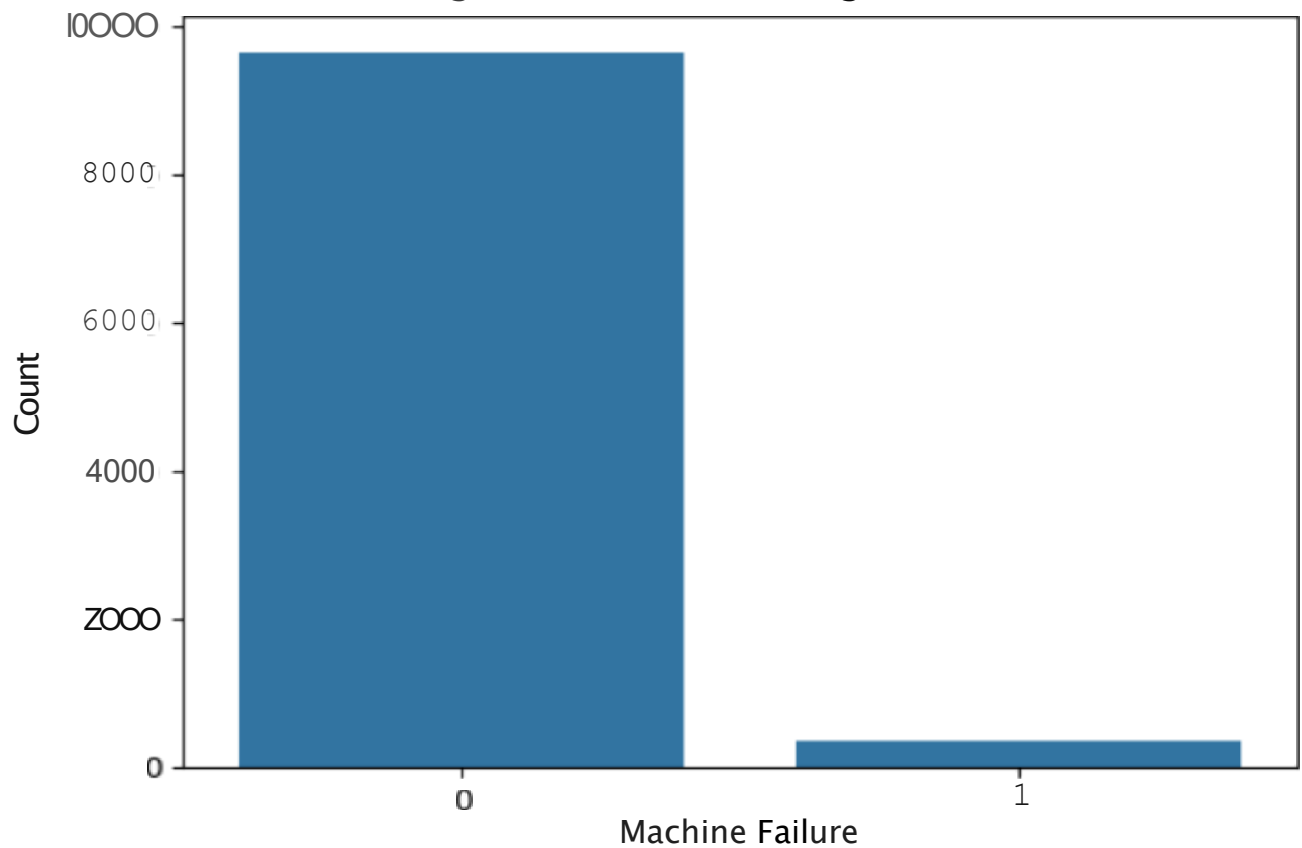


Fig 4: Count Plot of Target Variable



===== Fi O: VII VALMEU =====

1 [K] 90922.239693

3 Rotational speed 379.951018

oo r m

10 R h4F 1.003900

Fig 6: VIF of Features

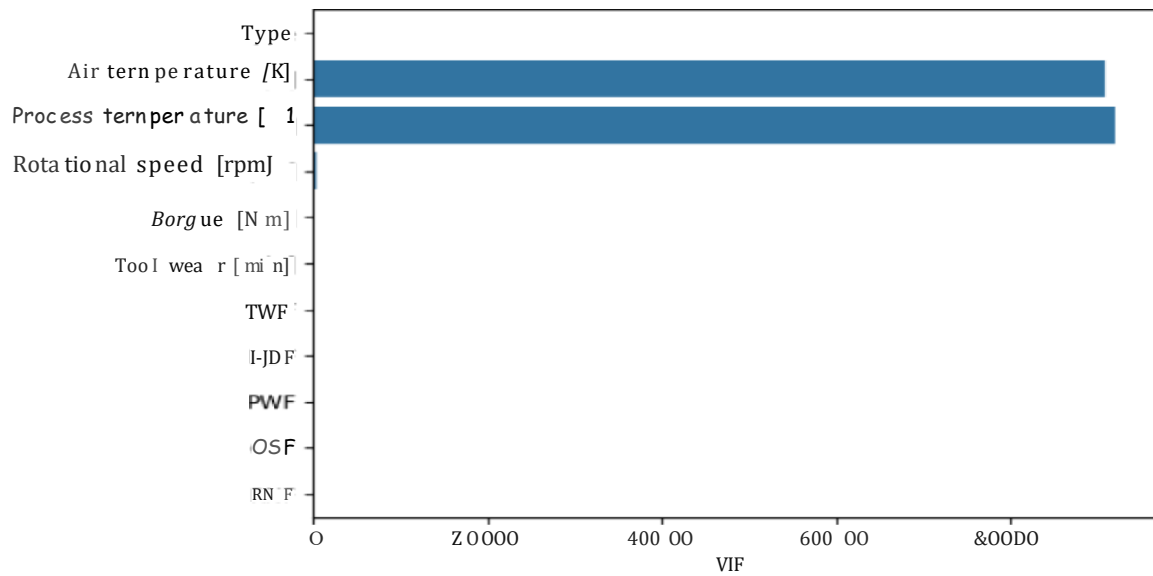
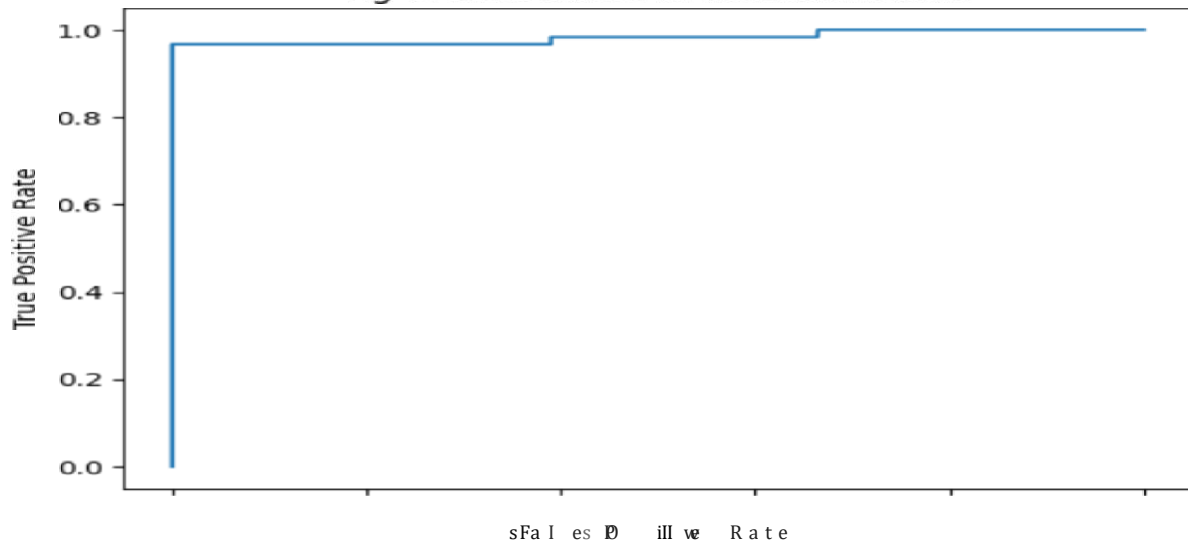


Fig 7: ROC Curve of XGBoost Model



7. Performance of Machine Learning and Deep Learning Models

Machine Learning Models:

- Faster training time
- Good baseline performance
- Less computational complexity
- However, limited in capturing complex patterns

Deep Learning Models:

- Better at handling complex and non-linear data
- Higher prediction accuracy
- Requires more data and computational power

Hybrid Model Advantage:

- Combines strengths of both approaches
- Achieves higher accuracy and robustness
- Optimized using evolutionary algorithms

8. Conclusion and Future Work

Conclusion:

The methodology successfully demonstrates the effectiveness of combining machine learning, deep learning, and evolutionary algorithms for predictive maintenance. The hybrid framework provides accurate and reliable predictions for anomaly detection and failure risk assessment.

The project successfully demonstrates the development of an intelligent neuro-evolutionary hybrid AI framework for anomaly detection and predictive maintenance. By combining neural networks with evolutionary optimization techniques, the system achieves improved prediction accuracy and robustness. The proposed model effectively identifies anomalies and predicts potential failures, enabling proactive maintenance strategies. This reduces operational downtime, minimizes maintenance costs, and enhances system reliability. Predictive maintenance is very useful. ML and DL help: Predict failures, Reduce cost, Improve efficiency. The **challenges**: Imbalanced data, High cost of sensors.

Future Work:

Future work can focus on integrating real-time data processing, deploying the model in large-scale industrial environments, and exploring advanced deep learning architectures. Better models and Low-cost solutions. **AI can make industries smarter and more efficient.**

The framework has significant potential for application in smart manufacturing and Industry 4.0 systems.

- Integration with real-time IoT systems

- Deployment in large-scale industrial environments
 - Use of advanced deep learning models like Transformer-based architectures
 - Improvement in computational efficiency
 - Development of automated maintenance decision systems
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